

Quiz 3 [mean 0, std 1]

① $P(X > 1.23)$

$$\begin{aligned}\frac{1.23 - 0}{1} &= 1.23 \\ &= 1 - \Phi(1.23) \\ &= 1 - 0.8907 \\ &= 0.1093 \# \end{aligned}$$

② $P(X < -0.25)$

$$\begin{aligned}\frac{-0.25 - 0}{1} &= 1 - \Phi(0.25) \\ &= 1 - 0.5987 \\ &= 0.4013 \# \end{aligned}$$

③ $P(-1.1 < X < 1.35)$

① $P(X > -1.1)$

$$\begin{aligned}\frac{-1.1 - 0}{1} &= 1 - \Phi(1.1) \\ &= 1 - 0.8643 \\ &= 0.1357 \end{aligned}$$

② $P(X < 1.35)$

$$\begin{aligned}\frac{1.35 - 0}{1} &= \Phi(1.35) \\ &= 0.9115 \end{aligned}$$

$$\therefore 0.9115 - 0.1357 = 0.7758 \#$$